

Manshay Verma

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SUMMARY

Backend and AI/ML engineer with hands-on experience building production backend systems (FastAPI, Django REST Framework), automating enterprise data workflows, and architecting agentic AI systems (LangGraph, LangChain, CrewAI, RAG pipelines). Comfortable working across cloud infrastructure (AWS), databases, and full-stack integrations, with a strong interest in ML/AI-driven backend engineering roles.

EXPERIENCE

Cubexo Software Solutions May 2025 - May 2026

Associate Software Engineer Indore, India

VI — Telecom Analytics & Backend Automation

- Built a large-scale telecom analytics and backend automation system using FastAPI, Pandas, and Shell Scripting on Linux-based server environments.
- Automated enterprise workflows for data ingestion, transformation, formula-based calculations, network analytics, and reporting pipelines, reducing manual processing time and improving operational efficiency.

Harman Studio — Multi-Tenant SaaS Platform

- Architected and developed a multi-tenant SaaS cloud infrastructure for professional audio studios using FastAPI, GraphQL, and AWS services.
- Built scalable backend systems to handle complex studio management workflows, supporting multiple independent studio tenants on shared infrastructure.

Harbinger — Enterprise Backend Services

- Developed scalable RESTful APIs using Django REST Framework for enterprise applications, implementing authentication, authorization, business logic, and asynchronous workflows while optimizing backend performance.
- Worked with MongoDB Atlas, AWS Lambda, Linux, GitHub, Jira, Confluence, and Sumo Logic to build reliable backend services, automate deployments, and improve infrastructure scalability.

PROJECTS

Autonomous Multi-Agent Research Assistant (LangGraph + RAG)

Technologies: LangGraph, LangChain, FastAPI, PostgreSQL, Redis, ChromaDB, Docker

- Architected a LangGraph-based multi-agent system (Supervisor, Researcher, Summarizer, Fact-Checker) that autonomously plans, retrieves, and verifies research information into a cited report, cutting manual research time by ~70% in testing.
- Implemented a RAG pipeline with ChromaDB and chunked embeddings, reducing factual hallucination rate by ~40% through an automated fact-checking agent loop.
- Built an async FastAPI backend orchestrating multi-agent workflows with SSE streaming, supporting 50+ concurrent research sessions.
- Designed a PostgreSQL + Redis architecture for persistent session state and short-term agent memory; containerized and deployed with Docker Compose on AWS EC2.

Customer Support Multi-Agent System (CrewAI)

Technologies: CrewAI, FastAPI, Celery, PostgreSQL, Redis, Docker

- Built a CrewAI-based customer support automation system with 3 specialized agents (Billing, Technical, Refund), achieving an 85% auto-resolution rate on a 500-ticket synthetic dataset.
- Designed an async ticket-processing pipeline using Celery and Redis, reducing average handling time by 45%.
- Implemented context-aware escalation logic that auto-generates handoff summaries for human agents, cutting escalation resolution time by 30%.

Personal Knowledge Base Agent (RAG + Long-Term Memory)

Technologies: LangChain, Weaviate, FastAPI, PostgreSQL

- Built a personal RAG chatbot with long-term memory, storing summarized past conversations as embeddings, improving multi-session context recall by 50%.
- Implemented multi-format document ingestion (PDF/DOCX/MD) with semantic chunking, improving retrieval relevance by 25% over fixed-size chunking.

Multi-Agent Travel/Itinerary Planner (CrewAI)

Technologies: CrewAI, FastAPI, PostgreSQL, Google Maps API, OpenWeatherMap API

- Developed a CrewAI-based multi-agent trip planner (Planner, Budget, Logistics agents) that iteratively refines itineraries, keeping 95% of generated plans within user budget constraints.
- Integrated Google Maps Distance Matrix and OpenWeatherMap APIs to validate itinerary feasibility, reducing scheduling conflicts by 40%.

TECHNICAL SKILLS

Programming Languages: Python, C++

Frameworks and Tools: Django, Django REST Framework, FastAPI, Flask, GraphQL, TensorFlow, PyTorch, LangChain, LangGraph, CrewAI, Celery, Docker

Concepts: Microservices Architecture, RESTful APIs, CI/CD, Machine Learning, Agentic AI, RAG Pipelines, Agile Development

Data & Analysis: Pandas, NumPy, SQL

Databases: PostgreSQL, SQLite, MongoDB, MySQL, ChromaDB, Weaviate

Cloud & Tools: AWS (EC2, S3, Lambda, RDS, Aurora, DynamoDB, VPC, IAM, Route 53, CloudFormation, ELB, Auto Scaling, CloudWatch, CloudFront, Athena), Linux, Sumo Logic, Jira, Confluence

Reporting & Visualization: Excel (formulas, pivot tables), Plotly Dash, Streamlit

EDUCATION

Devi Ahilya Vishwavidyalaya, Indore June 2023 - April 2025

Master of Technology in Computer Science

Relevant Coursework: Advanced Algorithms, Machine Learning & Artificial Intelligence, Distributed Systems, Cyber Security, Artificial Intelligence (Python), Data Processing & Web Development (Django, Flask).

Sri Aurobindo Institute of Technology, Indore March 2019 - May 2023

Bachelor of Technology in Computer Science

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Data Analytics & Cloud Computing, AWS.

CERTIFICATIONS

- Ultimate AWS Certified Solutions Architect Associate — Udemy (Stephane Maarek)
- The Complete Python Developer — Udemy (Andrei Neagoie)
- Complete A.I. & Machine Learning, Data Science Bootcamp — Udemy (Andrei Neagoie, Daniel Bourke)